

TABLE 1
Test Points

Test point	Details
TP0, TP5	0V (GND)
TP1, TP6	5V
TP2	Train of pulses when the controller transmits a signal
TP3	Train of pulses when the controller receives a signal
TP4	3.3V
TP7	Train of pulses when Arduino Nano receives a signal

with the values, one after the other, on the 7-segment display.

The GPS master clock is programmed to transmit only date and time to the slave clocks through an RF link. If the RF transmitter (TX1) is not used, it can be used as an independent GPS digital clock.

Slave clock. The slave clock is built on Arduino Nano board (BOARD1). A 16x2 character LCD (LCD1) is used to display time here. The LCD panel is connected on digital pins D2, D3, D4, D5, D6 and D7, and a 433MHz RF receiver (RX1) data pins are connected on digital pin 11 (D11) of the Arduino Nano board.

Data received from the master clock through the RF link includes date and time along with a password. Since this is an open signal on 433MHz RF link, the signal received at the receiver's end is checked with the password. If the password is correct, it will display the date and time exactly as the master clock, on the LCD.

Construction and testing

An actual-size, single-side PCB layout of the GPS master clock is shown in Fig. 6 and its component layout in Fig. 7.

An actual-size, single-side PCB layout of the GPS slave clock is shown in Fig. 8 and its component layout in Fig. 9.

The GPS module and buzzer in the master clock need to be disconnected while the sketch (Arduino software) is being uploaded onto the chip. Since our GPS module runs on 3.3V, a 3.3V

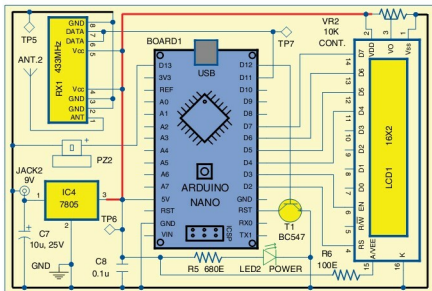


Fig. 5: Circuit diagram of the GPS slave clock

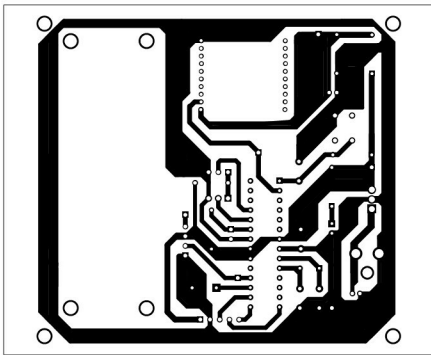


Fig. 6: Actual-size PCB layout of the GPS master clock

zener diode is used to step down from 5V to 3.3V with a 22-ohm resistor in between.

Use a suitable 5-pin connecting wire cable to connect DM1 with CON1 shown in the PCB. Switch on the GPS master clock by connecting to a 9V DC through JACK1 and vary VR1 to get the date and time on the display module (DM1). Now, switch on the slave clock by connecting to a 9V DC

through JACK2. You should be able to get the exact date and time as shown on DM1. If there is any display problem, check the voltages at various test points listed in Table 1.

Software

The master and slave programs (sketches) are written in Arduino programming language. Sketch for master clock is uploaded onto ATmega328